CHEM 427/527 · SESSION 15 · THURSDAY, NOVEMBER 19, 2026

Principal component analysis: which volcano did this obsidian come from?

Matt Bush, with teaching assistants Lucas Narisawa and Chris Weir

Drafted with assistance by Claude Code. Reviewed and revised by Matt.

BY THE END OF THIS SESSION YOU CAN

Five things to do with ten elements and a few components

1. Choose between centering and autoscaling from the variables’ units and variances, and show what the choice does to the first component and to a call.
2. Read a scores plot and its loadings in chemical terms, and tell a component that measures composition from one that measures how the sample was measured.
3. Choose the number of components from the scree plot and from how often held-out references are rejected, and say what the number does to a call.
4. Decide whether an unknown belongs to a reference class from its two distances to the class’s model and their limits, including “neither” when the data say so.
5. Review your notebook in a fresh context, check each finding against the cell it names before you act on it, and record one you fixed or rejected.

LAPTOPS, TO MINUTE 2

Run git pull; session 15 appears

In the terminal

```
git pull
```

Nothing is downloaded today: the data file is in the repository.

What appears

The folder S15-pca, with notes.md, activity.ipynb, figures/ and data/ holding obsidian_data.xlsx, assignments.csv and README.md.

Tuesday’s session 14 notebook is due at 1:30 pm today. Today’s is due at 1:30 pm on Tuesday, November 24.

THE DECISION

Which volcano did this obsidian come from?

Who decides

An archaeologist finds a flake of obsidian at a prehistoric site and decides which source it came from: the Slovak sources of Carpathian 1, the Hungarian sources of Carpathian 2, or neither. The evidence is an ED-XRF analysis of ten elements in the flake, set against analyses of samples from the sources.

Wrong source

The site’s supply is drawn to the wrong side of the hills, and a regional reading of who used which source moves with it.

Called neither

A link the data carry goes unpublished.

Real data: a Zenodo deposit by Petřík and Prokeš

One sheet of 246 ED-XRF analyses of Carpathian obsidian. After the filters, 53 reference analyses survive, 30 Slovak, 20 Hungarian and 3 Ukrainian, and 122 artefact analyses of 92 names. You have one name.

FROM HELD-OUT STATES TO A HELD-OUT REFERENCE

How would you check that a model of 30 Slovak analyses accepts a Slovak piece it has not seen?

Session 14

A correction was chosen by its error on states left out of the fit, because the error on its own rows flatters it.

Today

A model of one source is fitted to that source’s 30 reference analyses, and it accepts or rejects a new analysis by two limits.

The question

Which test tells you how often the model turns away its own kind?

Testing the model on the same 30 analyses is the flattering test session 14 set aside.

THE ANSWER, AND THE NUMBER OF COMPONENTS

Hold each reference out, fit without it, and count the rejections

Share of the 50 Slovak and Hungarian references rejected by their own source's model

Scaling	k = 1	k = 2	k = 3
Centered	20%	**16%**	18%
Autoscaled	20%	**16%**	26%

What the table says

Two components reject the fewest under either scaling, so the notebook sets K = 2.

The limits promise one rejection in twenty. With a few dozen references they give one in six, so a "neither" is read knowing that.

Too few components leave real composition in the residual; too many model the reference set's own scatter. The held-out count finds the middle.

LIVE: THE REFERENCES, CENTERED ONLY

Is barium the chemistry, or the biggest number?

Two variances

Element	Variance over the 53 references
Ba, in ppm	22,655 ppm²
CaO, in wt%	0.114 wt%²

A variance has the units of the column squared, so the unit decides which column is widest.

The first component, centered only

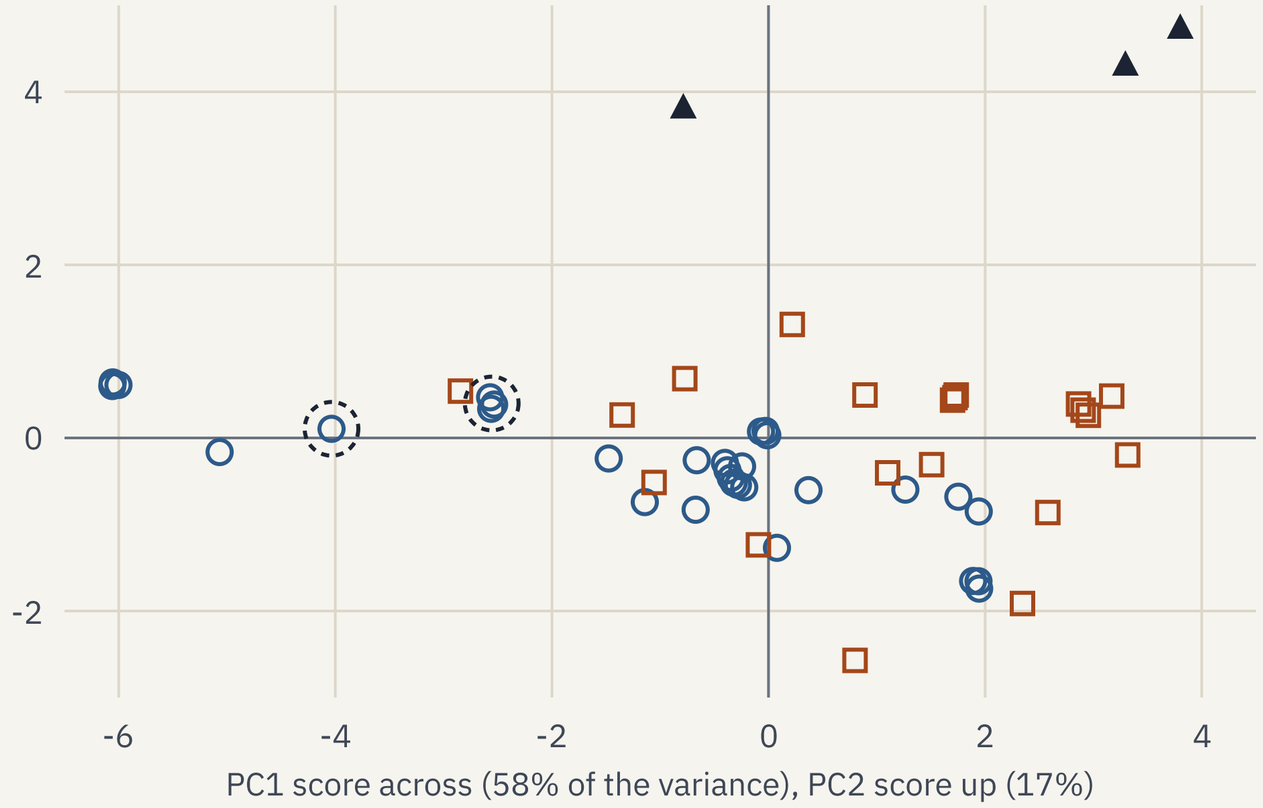
It carries 91% of the variance, and its loading on barium is +0.99. The other nine elements weigh 0.11 or less.

The component is barium in parts per million, not the chemistry of the sources.

Autoscaling divides each centered column by its standard deviation, so every element enters with variance one. The next slide shows what the components measure then.

LIVE: THE REFERENCES, AUTOSCALED

Which component does the measurement move?



Slovak, blue circles: mean PC1 −1.00.
Hungarian, orange squares: mean PC1 +1.18.
Ukrainian, black triangles: alone at PC2 +4.3.

Every PC1 loading is positive, 0.14 to 0.36: PC1 measures how much of everything a flake returned. PC2 sets Sr, CaO, Fe_2O_3 and V against Rb, Th, Ba, Al_2O_3, K_2O and SiO_2: a ratio.

The dashed rings are one Slovak reference measured in both campaigns: PC1 −2.56 in one and −4.03 in the other, PC2 +0.40 and +0.11. The measurement moves a flake along PC1 and leaves PC2 alone.

The Slovak and Hungarian means differ along PC1 too, so the call is made by the limits, not by eye.

ASSIGNMENTS

One artefact each, by student number

#	Artefact	#	Artefact	#	Artefact	#	Artefact
1	Kulna vr3-4	8	Moravany4	15	Bobrov1-1	21	Milovice 88 c 17
2	8-KosiceBarcaI	9	3-Tibava	16	NovaDedina2	22	Moravany8
3	NovaDedina6b	10	Cejkov7-acme-artefakt	17	StrSk-66.85c	23	neslovice-mikulasek
4	JarosovII	11	Moravany3	18	Koldin-3	24	StrSk-66.85
5	Bobrov1-3	12	StrSk-66.85-2-3	19	Miskovice-kontrol	25	NovaDedina2b
6	NitraIII-3	13	StrSk-66.85-2-2	20	NitraIII-2	26	Bobrov1-2
7	UH Sady	14	Moravany5				

Your artefact is beside your student number from session 4 in data/assignments.csv. In Parameters, set **ARTEFACT** to the name exactly as written, with its spaces, hyphens and periods, such as ARTEFACT = "Moravany4", with the quotes, and run the Parameters cell. Section 1 is the worked example on the references; section 2 is your artefact; section 3 makes the same call centered and autoscaled.

AI PRACTICE: A FRESH-CONTEXT REVIEW, NEW TODAY

A conversation that has never seen your notebook reads the file

How

Save the notebook, so the file holds your outputs. Open the Command Palette and run **Claude Code: Open in New Tab**. Type the review prompt from section 3 as written.

It takes one to two minutes and costs five to ten questions' worth of your usage limit: once per notebook, after section 3.

What comes back

A numbered list of findings of four kinds: a cell that would stop a run, a number in a markdown cell that disagrees with its output, a figure without units or its sentence, and a choice made outside Parameters.

Each names a cell by an id stored in the file, which VS Code does not show: find the cell by the words the finding quotes.

Open the cell before you act. A finding you reject with a reason is as good an entry as one you fix. Decline any offer to edit or run. The three lines of your prompt record count toward the execution point.

WHAT THE COURSE'S TRIALS OF THE REVIEW FOUND

Three findings to fix, two to reject with a reason

Right, by cell

- A TASK 2 cell that scaled the artefact by its own row, with the given cell's warning line quoted and the fix named.
- A plot sentence whose two distances were the wrong way round, with the printed line quoted against it.
- An Explain that called PC2 "the source" when the printed means put the two sources apart on PC1.

Unconfirmed, not wrong

- Every number a sentence took from a plot: the reviewer cannot see images. The plot is the answer.
- A neighbor's numbers in the pair Explain: they are in another notebook. The neighbor's output is the answer.

A finding you reject after opening the cell, with what it shows, is a grounded entry.

The review found nothing that would stop a run, and it never suggests wording for the judgment cell: it says only whether each part cites a number in the output.

THE JUDGMENT CELL, SECTION 4

Your source call, and one entry

(a) Your answer

- Which source did your artefact’s obsidian come from: CI, CII or neither?
- Your autoscaled T^2 and Q ratios to each source’s limits, your centered call, and whether one or three components would move the call
- What a wrong call does to the reading of the site’s supply

(b) One judgment entry

- Your prompt record from the review is the source.
- What you did: fixed, rejected, or verified
- The evidence: what the cell the finding named shows

What would you report for an artefact that is Slovak when the elements are centered and Hungarian when they are autoscaled?

TO SUBMIT

Restart, Run All, save, upload by 1:30 pm on Tuesday, November 24

Scored 0 to 3

Execution scores 1 when every cell runs after a restart and the notebook holds your prompt record from the review. The judgment cell scores 0 to 2 by the rubric in the syllabus.

Submissions are individual

Work beside a neighbor for the pair Explain cell; each of you submits your own activity.ipynb to the Session 15 assignment on Canvas.

The third hour is an office hour. Lucas grades session 15 and returns it by Tuesday, December 1.